

IS 507 Final Project

“Assessing Economics Factors in Corn Futures Price”

Member list: Sutthana Koo-anupong(sk188) / Yi-Han Huang(yihan18)

YUN CHIAO CHENG(yc87) / Zih-Yi Cao(zihyic2)

1. Abstract

The volatility of corn futures prices poses significant economic challenges, particularly in major agricultural regions like Illinois. This study investigates the influence of macroeconomic factors—such as the Consumer Price Index (CPI), oil prices, and the S&P 500 Index—and categorical corn production levels (Harvest_Level) on corn price fluctuations. Using historical trading and macroeconomic data, we developed and compared multiple regression and machine learning models to predict corn futures prices and identify key drivers of price variability.

Linear regression served as the baseline model, achieving an R^2 of 0.9434 but with high Mean Squared Error (MSE), indicating potential limitations like multicollinearity. Ridge and lasso regression introduced regularization to mitigate these issues, demonstrating the best performance among linear models (MSE: 1781, R^2 : 0.945) by effectively selecting relevant features. Non-linear methods, including Random Forest and gradient-boosting regressors, were employed to capture complex relationships. Random Forest outperformed linear models with an MSE of 1583 and R^2 of 0.95, highlighting its robustness and suitability for non-linear dependencies. In contrast, Gradient Boosting underperformed (MSE: 2147, R^2 : 0.933), underscoring the importance of hyperparameter tuning.

This comprehensive analysis underscores the significant roles of CPI, oil prices, and export indices in influencing corn prices. Feature importance analysis revealed the limited predictive power of the S&P 500 Index and Harvest_Level in the presence of other macroeconomic factors. Our findings demonstrate the value of combining linear and non-linear models to improve predictions and offer insights for stakeholders seeking to mitigate the economic impacts of corn price volatility.

2. Hypothesis

- 1) Changes in macroeconomic factors, such as the Consumer Price Index (CPI), export index, and oil prices, significantly influence corn futures price fluctuations.
- 2) Corn production levels, represented by the categorical variable Harvest_Level, have a measurable impact on corn futures prices, with higher production levels contributing to lower prices due to increased supply.

- 3) Returns on the S&P 500 Index indirectly influence corn futures prices, as changes in broader market conditions affect commodity investments.

3. Background

The corn price fluctuation is a critical concern for many stakeholders, especially in Illinois, one of the leading corn-producing states in the United States. According to the USDA, corn is the primary crop in the U.S., and price fluctuations can significantly impact farmers, food producers, and consumers alike. Understanding and predicting these fluctuations is essential for assessing their broader economic implications.

This project investigates the economic factors influencing corn futures prices. We collected and analyzed a diverse set of data, including historical corn futures trading data and macroeconomic indicators such as the Consumer Price Index (CPI), the Corn Export Index, U.S. Corn Export Index, oil prices, and corn production levels.

To enhance the analysis, we applied data preprocessing techniques. For instance, we transformed the corn production amounts into a categorical variable, `Harvest_Level`, scaled from 1 to 5, to evaluate its influence in a non-linear context. Additionally, we calculated the return of the S&P 500 Index, a major stock market indicator, to infer its potential relationship with corn price movements.

With this preprocessed data and feature engineering, we developed a regression model to predict corn prices and assess the relative impact of various economic factors. This model aims to provide insights into the drivers of corn price volatility and its implications for key stakeholders.

4. Proposed Method

To analyze the economic factors influencing corn futures prices and develop an effective predictive model, we implemented a multi-phase approach utilizing various regression and machine learning techniques.

In the initial phase, we employed a **linear regression model** to establish a baseline understanding of the relationships between the selected features and corn prices. However, recognizing the potential limitations of linear regression, such as overfitting or multicollinearity, we explored **ridge regression** and **lasso regression** to address these issues. Both methods include a regularization term that penalizes overly complex models, with ridge regression focusing on shrinking coefficients and lasso regression allowing for variable selection.

To optimize the performance of ridge and lasso regression, we performed **cross-validation** to determine the best value for the regularization parameter (α), ensuring that the models generalize well to unseen data.

In the subsequent phase, to address potential non-linear relationships and interactions that linear models might fail to capture, we explored machine learning techniques, including:

- **Random Forest Regression:** A tree-based ensemble method that can handle complex, non-linear interactions and automatically select important features.
- **Gradient Boosting Regression:** An iterative, ensemble learning method that builds models sequentially, optimizing performance by minimizing prediction errors.

Through this iterative approach, we aim to compare the performance of linear and non-linear models in predicting corn futures prices, selecting the best model based on accuracy, interpretability, and robustness.

5. Results

1) Linear Regression

In the initial phase of our analysis, we implemented a simple linear regression model to predict corn futures prices using the selected features. The model's performance was evaluated using two key metrics: Mean Squared Error (MSE) and R^2 .

The model achieved an MSE of **1833.27**, and an R^2 value of **0.9434** demonstrates that the linear regression model explains approximately **94.34%** of the variance in corn futures prices.

These results suggest that the linear regression model performs well in capturing the overall relationship between the features and corn price fluctuations. However, the relatively high MSE indicates that there is room for improvement in terms of predictive accuracy. To address potential limitations such as multicollinearity and to further optimize the model, we proceeded to test ridge and lasso regression.

2) Ridge and Lasso Regression

The high **Mean Squared Error (MSE)** observed in the linear regression model prompts the use of **ridge** and **lasso regression**. A high MSE indicates that the linear model is not fitting the data well, potentially due to overfitting, multicollinearity, or a lack of regularization. In this context, we apply ridge and lasso regression to incorporate regularization techniques that help improve model performance. By applying ridge and lasso regression, we aim to reduce the high MSE and enhance

the generalization ability of the model. However, the results from running **ridge** and **lasso regression** after **linear regression** only show slight improvements in model performance with regards to MSE and R^2 . While ridge regression performs almost identically to linear regression in terms of both MSE and R^2 , suggesting that its effect on model performance is minimal in this case, lasso regression achieves a better performance, with the lowest MSE (1781) and a slight improvement in R^2 (0.945). This indicates that lasso's ability to perform feature selection and shrink some coefficients to zero has improved the model's predictive accuracy. By removing less relevant features, the lasso model is more focused and generalizes better, leading to improved performance over both the linear and ridge models.

3) Comparison of Linear Regression Coefficients

Feature/Model	Simple Linear	Ridge Regression	Lasso Regression
Harvest Level	-2.1778	-2.1135	-0.0000
Temperature	-0.4697	-0.4708	-0.4734
CPI	0.8426	0.8417	0.8009
Export Index	1.6226	1.6230	1.6449
Oil Price	1.9249	1.9242	1.8537
S&P 500 Return	-2.6851	-0.1019	-0.0000

- A. **Oil Price** and **Export Index** exhibit positive coefficients on the three models, indicating a direct relationship with corn futures prices. As these values increase, corn prices are likely to rise. This aligns with economic expectations, as higher oil prices can increase production and transportation costs, while stronger export indices reflect increased demand.
- B. **CPI** also has a positive coefficient, suggesting that inflationary pressures are associated with higher corn prices, potentially due to broader cost increases in agriculture and commodities.
- C. **Temperature (Temp)** has a modest negative coefficient, indicating that higher temperatures might slightly reduce corn prices, possibly due to their impact on growing conditions or regional production.
- D. The differences in **S&P 500 Return** and **Harvest Level** across the three models highlight the varying treatment of feature importance. For **S&P 500 Return**, linear regression assigns a strong negative coefficient (-2.69), suggesting a significant inverse relationship with corn prices. However, ridge regression drastically reduces this coefficient to -0.10, indicating that its influence may be overstated.

due to multicollinearity or noise. Lasso regression further eliminates this feature altogether (coefficient=0), implying it has limited predictive power in the presence of other factors. On the other hand, **Harvest Level** consistently demonstrates a negative relationship in linear (-2.18) and ridge regression (-2.11), reflecting its expected role in lowering corn prices with higher production levels. However, lasso regression excludes this feature entirely, suggesting its impact may overlap with other predictors such as **Oil Price** or **Export Index**. These patterns underscore the need for using multiple regression methods to assess feature relevance, control multicollinearity, and identify potential redundancy among variables.

4) Random Forest Regressor

After employing linear, ridge, and lasso regression models, we observed limitations in capturing complex, non-linear relationships among the features. We chose Random Forest as the model for predicting crop prices because it effectively handles complex relationships between features and the target variable. Combining multiple decision trees makes Random Forest robust to noise and outliers, and averaging predictions across trees helps reduce overfitting.

Performance-wise, Random Forest achieves an MSE of 1583, lower than Linear Regression's MSE of 1833, indicating more minor prediction errors. Its R^2 score of 0.95 is also slightly higher than Linear Regression's 0.943, suggesting it explains more variance in crop prices. While the improvement over Linear Regression is not drastic, Random Forest's ability to handle nonlinear relationships and feature interactions makes it a better fit for this dataset, which likely contains complex dependencies.

5) Gradient Boosting Regressor

After trying the Gradient Boosting model, we observed that its Mean Squared Error (MSE) was even higher than that of the Random Forest model. Gradient Boosting Regressor builds trees one at a time, with each tree improving on the mistakes made by the previous ones. This step-by-step approach helps it find complex patterns in the data, making it ideal for datasets with complicated relationships between features and the target. However, to achieve optimal results, it is crucial to carefully adjust parameters such as the learning rate, number of trees, and tree depth. In this case, Gradient Boosting has the lowest R^2 score of 0.933 among all models, which explains the least variance in crop prices. Its MSE of 2147 is also the highest, indicating larger prediction errors than Random Forest (MSE 1583) and Linear Regression (MSE 1833). It failed to perform as expected, possibly due to overfitting, underfitting, or suboptimal hyperparameter settings. This outcome highlights the importance of

experimenting with different models and tuning their parameters to identify the most suitable approach for a given dataset.

Figure 5.1 Comparison Across Different Regressions

	Linear	Ridge	Lasso	Random Forest	Gradient Boosting
MSE	1833	1832	1781	1583	2147
R ²	0.943	0.943	0.945	0.95	0.933

6. Discussion

1. Hypothesis Evaluation

First hypothesis which stated that changes in macroeconomic factors significantly influence corn futures price fluctuations is supported by the findings. The analysis showed positive coefficients for both Oil Price, Export Index, and CPI across all regression models. Further investigation could focus on quantifying how much each of the factors influences corn prices through sensitivity analyses or simulations.

Second hypothesis which posited that corn production levels or Harvest_Level have an impact on corn futures prices by higher production levels will contribute to lower prices yielded mixed results. While linear and ridge regressions supported the hypothesis, the lasso regression excluded this feature. This suggests that the relationship between production levels and prices is more complex and may depend on interactions with other factors. Future research should explore how these combined factors influence corn prices.

For the last hypothesis which proposed that the S&P 500 index would indirectly influence corn prices, our findings also showed mixed results. The varying coefficients across the models indicated potential issues with this hypothesis and we might need to revise it. Additionally, we could explore alternative market conditions indicators and consider potential lagged effects on corn prices.

2. Limitation and Improvement Area

A. Scope of macroeconomic factors: the project focused on a specific set of these factors. Future research should expand the analysis to include additional variables such as interest rates, currency exchange rates, government policies related to agriculture, global supply and demand dynamics and potential lagged effects to gain a more comprehensive understanding of factors driving corn price fluctuations.

B. Reliance on limited historical data: due to the limitation of our dataset which solely relied on monthly historical data, it might affect the model's prediction. If possible, we suggest incorporating real-time daily data and exploring advanced predictive modeling techniques to improve accuracy

C. Optimizing Gradient Boosting: the gradient boosting model underperformed in this study which indicates the importance of model selection and tuning. Future research should focus on optimizing model parameters and addressing data noise and exploring techniques like hyperparameter tuning and data cleaning to enhance the model's predictive capabilities.

7. Conclusion

The project successfully developed a model to predict corn futures prices by incorporating macroeconomic factors, corn production levels, and broader market conditions. CPI and oil prices were identified as significant factors influencing corn prices, as anticipated, while the S&P 500 had an indirect effect with a less pronounced impact than expected. Although the mean squared error (MSE) remains high, the project provided valuable insights into how various features impact corn futures prices to different extents. These findings serve as a foundation to guide future efforts in enhancing the model by employing advanced machine learning techniques and refining feature selection. This project demonstrates the practical application of machine learning concepts and theories covered in class and lays the groundwork for more accurate and reliable price predictions, benefiting stakeholders in the agricultural sector.

8. References

Economic Research Service, U.S. Department of Agriculture. (2024). U.S. Agricultural Trade Data Update. *U.S. Department of Agriculture*.
<https://www.ers.usda.gov/data-products/foreign-agricultural-trade-of-the-united-states-fatus/u-s-agricultural-trade-data-update/>

Investing.com. (2024). S&P 500 (SPX). *Investing.com*.
<https://www.investing.com/indices/us-spx-500-historical-data>

U.S. Energy Information Administration. (2024, December 12). Petroleum & Other Liquids. *U.S. Energy Information Administration*.
<https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=F0020173&f=M>

U.S. Bureau of Labor Statistics. (2024). Consumer Price Index. *U.S. Bureau of Labor Statistics*.
<https://data.bls.gov/pdq/SurveyOutputServlet>

National Weather Service. (2024). NOWData - NOAA Online Weather Data. *National Weather Service*.
<https://www.weather.gov/wrh/Climate?wfo=lot>